Proposal Calibration and Fold Consensus for Solar Filament Instances

A Solution to the Solar Filament Segmentation Challenge 2026

Jean-Louis Roy
jeanlouisroyroy@gmail.com

ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Our submitted V23 model combines five YOLO11s proposal detectors with crop refiners that learn mask correction and proposal acceptance. Three folds use an existence- and mask-quality-aware refiner (V3); two retain the simpler V2 refiner. A one-to-one association across folds retains instances supported by at least three models and selects an actual member mask as the medoid. Validation groups all annotations of a physical observation into the same fold and scores each annotator independently. The recovered experiment record reports out-of-fold panoptic quality (PQ) of 0.43520 for this hybrid; its verified public leaderboard score is 0.39. We provide model checkpoints, source, provenance, and the reference-submission identity. Further high-resolution and zero-shot refinement experiments did not justify replacing this reference. Reported historical validation values are distinguished from the checks performed while preparing the release.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

We focus on the reproducible V23 submission: a detector, proposal-refinement, and fold-consensus pipeline. It extends the public YOLO-plus-refiner baseline of ektarr [3] through proposal rejection, quality prediction, uncertainty-aware targets, and controlled fusion. The submitted hybrid contains V3 refiners for folds 0–2 and V2 refiners for folds 3–4. The backbone architectures and pretrained representations are existing components; our contribution is their task-specific training and selection pipeline.

The training annotations may give the same physical image several image identifiers, one per annotator. Treating these identifiers as independent observations would permit identical pixels in both training and validation. Our split unit is the image filename. The final system uses only the competition's official training labels for task-specific supervision. It does not consume labels for the competition test observations from another dataset release.

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

2.1 Data and supervision

The competition training split contains 707 physical JPEG images represented by 1,154 annotator records; the test split has 180 images. We retain the original 2048×2048 coordinate system when rasterizing masks and producing predictions. MAGFiLO provides instance polygons, spines, and chirality; its independent annotations may disagree about instance extent or whether a feature is one object or several [1]. The submitted model predicts a single filament category rather than chirality.

For training-target construction, annotations of the same physical image are associated by mask IoU. A representative annotator record is selected by maximum average semantic-mask agreement. Hungarian assignment matches other records to representatives, using an IoU threshold of 0.15; unmatched annotations create additional groups. The catalog retains individual member masks and annotator support, rather than assuming that their union is an additional observation. This makes disagreement available to proposal training without changing the independent annotations used for evaluation.

The saved five-fold manifest is reused throughout the model lineage. The source's fold builder balances physical-image counts while stratifying by observing-site suffix and instance-count quantile. Fold-specific detectors and refiners exclude the corresponding validation filenames from training. These are physical-image holdouts; they do not establish a temporal, site-held-out, or prospective evaluation.

2.2 Proposal generation and crop refinement

Each fold uses a YOLO11s segmentation detector [7] inherited from our trained baseline. The submitted inference path runs the detector at 1024-pixel input size with confidence 0.05, box-NMS IoU 0.80, class-agnostic NMS, and native-size masks. This permissive candidate generation is followed by mask-level NMS at IoU 0.45. V2 retains detector scores of at least 0.20 and at most 20 seeds; V3 retains scores of at least 0.05 and at most 24 seeds. Thus the refiner, rather than only the detector confidence, decides whether a proposal should survive.

A square crop surrounds each seed with a context multiplier of 1.8 and minimum side length of 128 native pixels. Crops are resized to 384×384 for V2 or 448×448 for V3. Three input channels contain the grayscale image, its local contrast-enhanced version, and the binary seed. CLAHE uses clip limit 2.0 and an 8×8 tile grid. The encoder is ResNet-18 [4], followed by a U-shaped decoder with skip connections. The final V23 refiners output one mask channel; they do not output a separate spine channel.

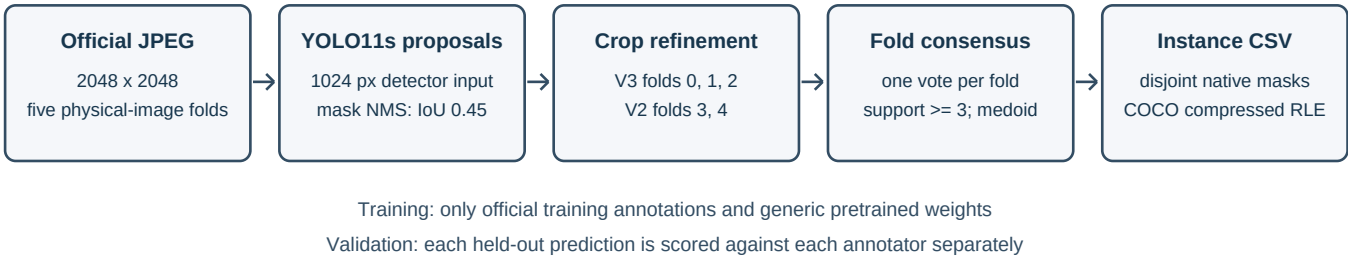


Figure 1: End-to-end V23 inference pipeline. Each fold supplies at most one vote to a consensus instance. Training-label construction and annotator-wise validation are separate operations.

V2 adds a pooled keep/reject head to the refiner. Its supervised examples include detector proposals and generated seed perturbations, with negative proposals teaching both rejection and suppression of mask foreground. The positive mask objective combines boundary-weighted binary cross-entropy (weight 0.45) and soft Dice (0.55). Proposal acceptance uses binary cross-entropy. V2 is initialized from the corresponding baseline refiner and trained for eight epochs with AdamW, learning rate 10^{-4}, weight decay 10^{-4}, and cosine scheduling. Training augmentation includes rotations, flips, intensity gain, and bias changes.

V3 starts from V2 and adds an existence correction conditioned on proposal metadata, together with a predicted-mask IoU head. Existence targets express annotator support for the matched object. The quality head learns the IoU of the current thresholded prediction, rather than the seed's original IoU. Its positive mask objective combines boundary-weighted cross-entropy, Dice, and a soft-skeleton term with weights 0.40, 0.50, and 0.10. The skeleton term follows the soft-clDice principle [6]; it encourages connectivity but does not prove correct filament topology. The total loss also includes negative-mask suppression (0.04), existence cross-entropy (0.35), and positive-mask IoU regression (0.25).

V3 uses AdamW with learning rates 5×10^{-5} for inherited parameters and 2×10^{-4} for new heads, weight decay 10^{-4}, cosine scheduling, mixed precision, and gradient clipping at 5. The resumed training retained the original six-epoch fold 0 and trained folds 1–4 for four epochs. Final hybrid inference uses only V3 folds 0–2. These details describe the saved training configuration, not a newly repeated training run.

2.3 Consensus and submission encoding

Table 1 lists the acceptance policy; e denotes predicted existence and $\hat{u}$ predicted mask IoU. Crop probabilities are restored to native coordinates before mask comparison. For each physical test image, predictions from the five folds are clustered using one-to-one Hungarian assignment at mask IoU 0.20. A fold cannot contribute duplicate votes to the same cluster. The support-three policy retains clusters with at least three distinct folds.

The medoid is the member mask with greatest average pairwise overlap within its cluster. Selecting a member preserves a model's contour instead of constructing a new boundary by pixel voting.

Table 1: Parameters of the final V23 inference path.

Parameter	V2 (folds 3–4)	V3 (folds 0–2)
Crop side	384	448
Detector-score cutoff	0.20	0.05
Maximum selected seeds	20	24
Acceptance cutoff	$q \geq 0.50$	$e \geq 0.55$
Expected-quality cutoff	—	$e\hat{u} \geq 0.30$
Mask-probability cutoff	0.50	0.50
Flip averaging	Four views	Identity + horizontal

Masks are ordered by support and score to resolve overlapping pixels. A cleaned mask must retain at least 96 pixels and 65% of its original area. If consensus leaves no candidate, the implementation falls back to the strongest available cluster; the recovered test run used this fallback on one of 180 images. This exception is part of the submitted algorithm.

The final CSV contains 1,173 instances, or 6.52 per test image, in two columns: `filament_id` and `segmentation_rle`. Masks are encoded with the COCO API as compressed RLE from Fortran-contiguous arrays. The output is checked for image coverage, valid identifiers, decodability, mask dimensions, and overlap. Figure 2 shows masks decoded directly from the archived submission, without test labels.

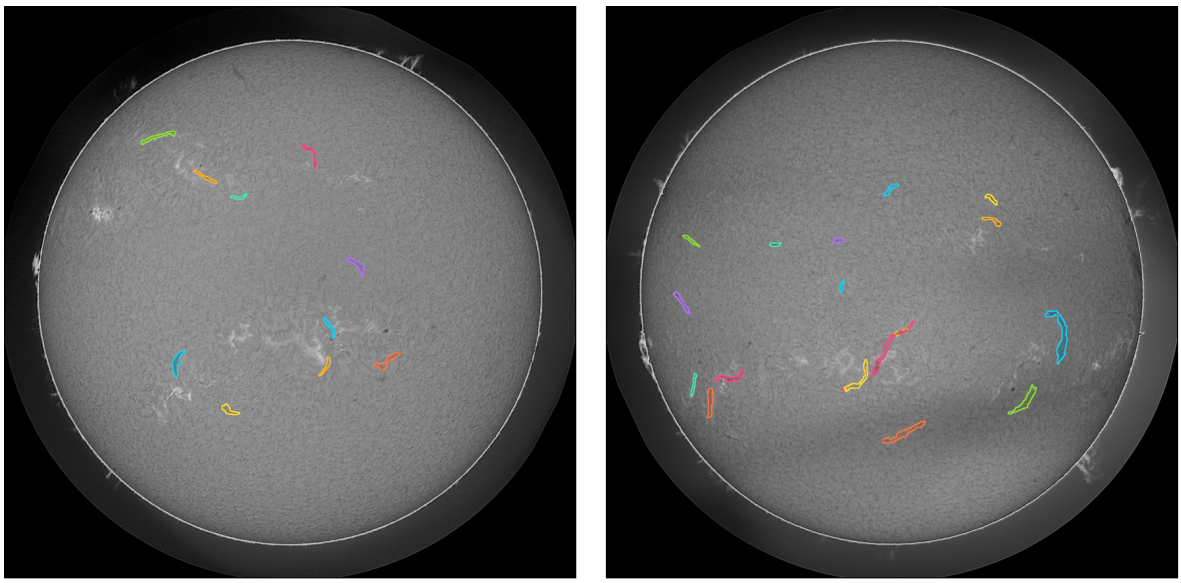
3 EVALUATION

3.1 Metric and evidence boundaries

We follow the competition's annotator-wise PQ convention, using strict IoU > 0.5 to identify a matched predicted and ground-truth instance. For matched pairs T, false positives F_P, and false negatives F_N, PQ [5] is

$$\mathrm{PQ} = \frac{\sum_{(g,p)\in T} \mathrm{IoU}(g,p)}{|T| + \frac{1}{2}|F_P| + \frac{1}{2}|F_N|}. \tag{1}$$

Each physical image's prediction is compared separately with every available annotator record. The numerator and counts are pooled across those records before taking the ratio; this is not the arithmetic mean of per-image PQ. Segmentation quality is mean



(a) 20110329082654Uh - 10 predicted instances (b) 20140519195834Ch - 17 predicted instances

Figure 2: Illustrative test predictions from the exact V23 reference CSV, overlaid on the official JPEGs. Different colors distinguish predicted instances. These images are qualitative examples, not a ground-truth comparison or evidence of per-image accuracy.

Table 2: Recovered reference results. OOF values are from the V23 hybrid-sweep record; public scores are verified Kaggle records.

Refiner composition	OOF PQ	Public	Instances
V2, all five folds	0.43119	0.39	1,273
V3, all five folds	0.43439	0.38	1,115
V23 hybrid 0–2 / 3–4	**0.43520**	**0.39**	1,173

IoU over matched pairs; recognition quality is the corresponding instance F1 factor. Consequently, a contour that falls just below the match threshold loses both its matched IoU contribution and its true-positive status.

All OOF values below are recovered historical experiment results, not fresh reruns performed for this report. The hybrid result is explicitly stored as a strict-OOF reference in the V23 output JSON. That output also records the test CSV's hash and counts, while Kaggle's retrieved submission history independently records its public score. We do not report a private leaderboard score or a confidence interval for V23 because neither is available in the recovered evidence. Threshold and architecture selection used validation experiments, so the selected OOF score should not be interpreted as an untouched final test estimate.

3.2 Ablations and unsuccessful extensions

Table 2 shows a small validation advantage for the hybrid over all-V3, without a public-score improvement over V2. An older V2 submission description reports OOF PQ 0.43383; the hybrid-sweep JSON instead records 0.43119 for its V2 reference. We preserve both provenance records and use the latter consistently in the table. The difference has not been resolved by a complete validation replay and should not be attributed to architecture alone. Switching hybrid fusion from medoid to strict pixel majority produced 1,174 instances and the same rounded public score, 0.39. Public scores reported to two decimals cannot resolve small differences.

A native high-resolution detector was also examined on the canonical fold-0 holdout: 141 physical images and 247 annotator records. Its best recovered PQ was 0.41757, compared with 0.44302 for the proposal-V3 reference on those same records. A nested five-fold ridge selector, using prediction-derived features to choose between these two outputs, achieved 0.44338. The recorded bootstrap 95% interval for its paired improvement was $[-0.00042, 0.00146]$. The apparent gain of 0.00035 therefore did not support promotion. Oracle selection by ground-truth performance reached 0.45514, but uses labels at selection time and is not a deployable model.

A separate zero-shot CascadePSP screen used 32 training-derived crops, half from dense seeds and half from YOLO seeds. The initial mean IoU was 0.50101; the best tested threshold gave 0.50367, a mean increase of 0.00267. Nine cases crossed upward through IoU 0.5 and six crossed downward. This small, selected crop screen is neither a full-image PQ evaluation nor a statistically established gain. The experiment's promotion gate failed; it did not read test images and was excluded from V23.

An attributed HDJojo control and its derivative scored 0.55. They are external artifacts, not V23; the originating checkpoint was not recovered, so they are excluded from our own reproducible result table.

3.3 Strengths, limitations, and reproducibility

The pipeline's main strengths are physical-image separation, proposal rejection informed by real detector errors, and explicit fold consensus. Its limitations are a detector-dependent candidate ceiling, reduced recall from support-three filtering, inconsistent human boundaries, and sensitivity to the hard IoU match threshold. The folds also share architecture and training data, so their agreement is not a calibrated probability of correctness. A useful next step is a complete replay of validation predictions with frozen settings, followed by a grouped uncertainty analysis. New models should pass this comparison before replacing the archived reference.

The public repository for this release is https://github.com/Jeanjean59800/solar-filament-segmentation-2026. The accompanying package contains the source notebook, five detector checkpoints, three V3 and two V2 refiner checkpoints, the reference-submission hash, and provenance manifests. Historical baseline logs identify Python 3.12.13, PyTorch 2.10.0+cu128, and Ultralytics 8.4.103. The original V23 four-variant inference sweep took 1.42 hours according to its recovered report; this is not a benchmark of a single-variant release command. The exact V23 CSV (Kaggle submission 55607242) has SHA-256 beginning `58b3a7c52c239d30`; the complete hash is included in the release manifest. Checkpoint recovery and output-file integrity checks support traceability; they do not substitute for a new end-to-end numerical reproduction.

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